

Chapter-1

Solutions

KEY IDEA

A solution is a homogeneous mixture of two or more substances. The composition can be varied within certain limits.

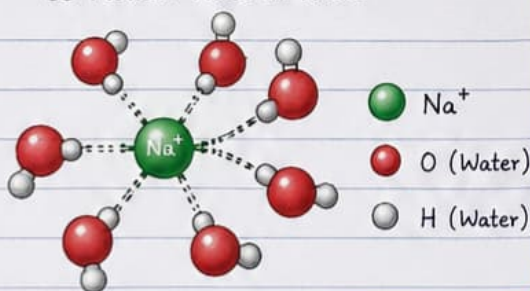
1.1 What is a Solution?

A solution is a homogeneous mixture in which the particles of the solute are uniformly distributed in the solvent.

Example:

- Salt (NaCl) dissolved in water
- Sugar dissolved in water

3D Model: NaCl in Water



1.2 Components of a Solution

- **Solute** → The substance that gets dissolved (usually in smaller amount)
- **Solvent** → The substance that dissolves the solute (usually in larger amount)
- **Solution** → The homogeneous mixture formed



e.g. NaCl (solute) + H_2O (solvent) → NaCl solution

1.3 Types of Solutions

On the basis of physical state:

- **Solid in Solid** → Alloys (e.g., Brass)
- **Solid in Liquid** → Salt in water
- **Liquid in Liquid** → Alcohol in water
- **Gas in Liquid** → CO_2 in water (soda water)
- **Gas in Gas** → Air (O_2 in N_2)

1.4 Concentration of Solutions

It tells how much solute is present in a given amount of solution or solvent.

① Mass % (w/w)

$$\text{Mass \%} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

② Volume % (v/v)

$$\text{Volume \%} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

③ Mass by Volume % (w/v)

$$\text{Mass by volume \%} = \frac{\text{Mass of solute}}{\text{Volume of solution (mL)}} \times 100$$

④ Mole Fraction (X_A)

$$X_A = \frac{\text{Number of moles of A}}{\text{Total moles of all components}}$$

Example 1: Calculate mass % of NaCl in a solution containing 15 g of NaCl in 150 g of water.

Solution: Mass of solute = 15 g, Mass of solvent = 150 g, Mass of solution = 165 g

$$\text{Mass \%} = \frac{15}{165} \times 100 = 9.09 \%$$

Quick Tip

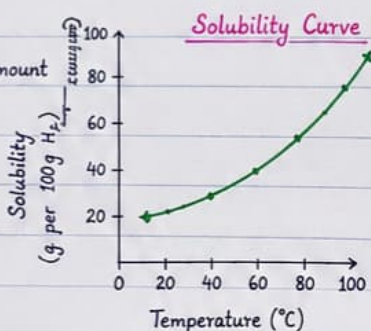
Mass % + Mass of solvent = Mass of solution

1.5 Solubility

Solubility is the maximum amount of solute that can dissolve in a given amount of solvent at a specific temperature to form a saturated solution.

Note

- Solubility generally ↑ with ↑ temperature (for solids in liquids)
- For gases in liquids, solubility ↓ with ↑ temperature.



Example:

At 20°C, solubility of KNO_3 in water is ~32 g per 100 g of water.

1.6 Factors Affecting Solubility

The solubility of a substance depends on the nature of solute & solvent, temperature and pressure.

① Nature of Solute & Solvent

- "Like dissolves like."
- Polar solute dissolves in polar solvent.
- Non-polar solute dissolves in non-polar solvent.

Examples:

- NaCl (ionic, polar) → soluble in water (polar)
- I₂ (non-polar) → soluble in CCl₄ (non-polar)
- I₂ (non-polar) → insoluble in water (polar)

③ Effect of Pressure (for gases)

- Solubility of a gas in liquid increases with increase in pressure.
- (Henry's Law) : At constant temperature, solubility of a gas is directly proportional to the pressure of the gas above the liquid.

$$S \propto P$$

S = Solubility of gas

P = Pressure of gas

k_H = Henry's constant

$$S = k_H P$$

② Effect of Temperature

- For most solid solutes in liquids, solubility increases with increase in temperature.
- For gases in liquids, solubility decreases with increase in temperature.

Examples:

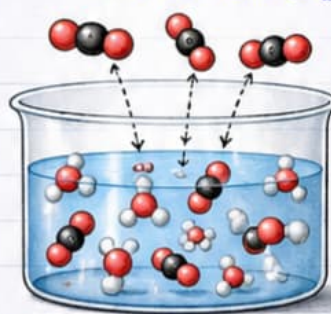
- Solubility of KNO₃ in water ↑ with temperature.
- Solubility of O₂ in water ↓ with temperature.

Why?

↑ T → more kinetic energy → forces of attraction between solute & solvent (in case of gases) are overcome → solubility of gases decreases.



3D Model: Gas (CO₂) in Water



- Red = Oxygen (O)
- Black = Carbon (C)
- White = Hydrogen (H)
- > = Dissolution of CO₂ in water

1.7 Expressing Concentration of Solutions

Concentration tells how much solute is present in a given amount (mass or volume) of solution.

① Mass % (w/w)

$$\text{Mass \%} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

Example:

10 g of NaCl in 90 g of water.

$$\text{Mass \%} = \frac{10}{100} \times 100 = 10\%$$

② Volume % (v/v)

$$\text{Volume \%} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

Example:

20 mL ethanol in 80 mL solution.

$$\text{Volume \%} = \frac{20}{100} \times 100 = 20\%$$

③ Mass by Volume % (w/v)

$$\text{Mass by volume \%} = \frac{\text{Mass of solute (g)}}{\text{Volume of solution (mL)}} \times 100$$

Example:

5 g NaCl in 100 mL solution.

$$\text{Mass by volume \%} = \frac{5}{100} \times 100 = 5\%$$

④ Mole Fraction (X_A)

$$X_A = \frac{\text{Number of moles of A}}{\text{Total moles of all components}}$$

Example:

2 mol NaCl + 3 mol H₂O

$$X_{\text{NaCl}} = \frac{2}{2+3} = \frac{2}{5} = 0.40$$

1.8 Types of Solutions (based on concentration)

Type	Description	Example	Particle View (3D Model)	
Dilute Solution	Contains small amount of solute.	A pinch of sugar in water		● Blue = Solute particle (e.g., Na⁺, sugar) ● Red = Oxygen ● White = Hydrogen
Concentrated Solution	Contains large amount of solute.	Saturated sugar solution		
Saturated Solution	Contains maximum amount of solute that can dissolve at a given temperature.	Saturated solution of KNO ₃ at 30°C		
Unsaturated Solution	Contains less solute than the saturated solution.	Salt water (less salt added)		

Extra solute remains undissolved (in equilibrium) with solution.

1.9 Methods of Expressing Concentration

① Mass % (w/w)

$$\text{Mass \%} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

Example:

10 g of sugar in 90 g of water.

$$\text{Mass \%} = \frac{10}{100} \times 100 = \boxed{10\%}$$

② Volume % (v/v)

$$\text{Volume \%} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

Example:

20 mL of ethanol in 80 mL solution.

$$\text{Volume \%} = \frac{20}{100} \times 100 = \boxed{20\%}$$

③ Mass by Volume % (w/v)

$$\text{Mass by volume \%} = \frac{\text{Mass of solute (g)}}{\text{Volume of solution (mL)}} \times 100$$

Example:

5 g of NaCl in 100 mL solution.

$$\text{Mass by volume \%} = \frac{5}{100} \times 100 = \boxed{5\%}$$

④ Molarity (M)

Molarity is the number of moles of solute present in 1 litre (1000 mL) of solution.

$$\text{Molarity (M)} = \frac{\text{Moles of solute}}{\text{Volume of solution (in L)}}$$



Example:

1 mole of NaCl dissolved in water to make 1 L of solution.

$$\text{Molarity} = \frac{1}{1} = \boxed{1 \text{ M}}$$

⑤ Molality (m)

Molality is the number of moles of solute present in 1 kg (1000 g) of solvent.

$$\text{Molality (m)} = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$$



Example:

2 moles of urea dissolved in 1 kg of water.

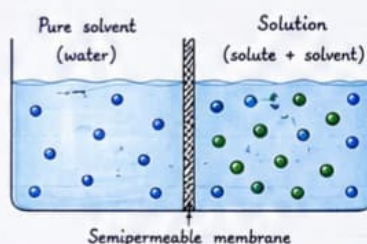
$$\text{Molality} = \frac{2}{1} = \boxed{2 \text{ m}}$$

Why do colligative properties depend only on number of particles?

☆ Colligative properties arise due to the presence of solute particles which affect the physical behavior of the solvent.

☆ More particles → greater effect. (regardless of whether the solute is ionic or molecular).

☆ Hence, they are called "colligative" (properties that "collectively" depend on number of particles).



Solvent moves from pure solvent → solution until osmotic pressure is balanced.

1.10 Colligative Properties

Colligative properties depend **only on the number of solute particles** present in the solution, not on the nature of the solute.

Property	Description	Effect of adding non-volatile solute
1. Relative lowering of vapour pressure (ΔP)	Decrease in vapour pressure of solvent.	Vapour pressure ↓
2. Elevation in boiling point (ΔT_b)	Increase in boiling point of solution.	Boiling point ↑
3. Depression in freezing point (ΔT_f)	Decrease in freezing point of solution.	Freezing point ↓
4. Osmotic pressure (π)	Pressure required to stop the flow of solvent through a semipermeable membrane.	Osmotic pressure ↑

1.11 Applications of Solutions in Daily Life

☆ **Electrolyte drinks:** Solutions of salts and sugars help in rehydration.

☆ **Medicines:** Many medicines are given in solution form for easy absorption (e.g., ORS, syrups).

☆ **Disinfectants:** Dettol, antiseptic solutions are used to kill germs.

☆ **Food industry:** Flavours, colours, soft drinks, sauces are all solutions.

☆ **Agriculture:** Fertilizers are dissolved in water and used as solutions.

☆ **Cleaning agents:** Soaps and detergents work as solutions.



Quick Revision

- ✓ Solution = Homogeneous mixture.
- ✓ Solute + Solvent → Solution
- ✓ Concentration tells amount of solute present.

- ✓ Molarity (M) = moles / litre of solution
- ✓ Molality (m) = moles / kg of solvent
- ✓ Colligative properties depend only on number of particles.

Remember

More particles → Greater effect (lowering vapour pressure, raising boiling point, etc.)



1.12 Important Points to Remember

- ① Solution is always **homogeneous**.
- ② Particle size in solutions is $< 1 \text{ nm}$.
- ③ Solutions **do not** scatter light (Tyndall effect is absent).
- ④ Solutions are **stable** and do not settle on standing.
- ⑤ We can **pass filter paper** through a solution.
- ⑥ Concentration can be expressed in **many ways**.
- ⑦ **Colligative properties** depend only on number of particles.
- ⑧ **Temperature** and **pressure** affect solubility.

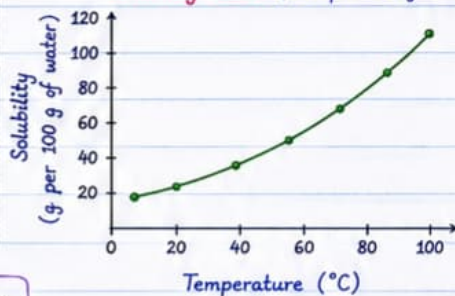
1.13 Solubility

Solubility is the maximum amount of solute that can dissolve in a given amount of solvent at a **specific temperature** to form a **saturated solution**.

Factors Affecting Solubility

1. Nature of solute & solvent
2. **Temperature**
3. **Pressure**

Solubility Curve (Example: KNO_3 in water)



From the curve, at 60°C solubility of KNO_3 in water is about 110 g per 100 g of water.

1.14 Saturated, Unsaturated & Supersaturated Solutions

Type	Description	How it looks	Example
Unsaturated	Contains less solute than the maximum amount that can dissolve at a given temperature.		Salt water with a spoon of salt that dissolves completely.
Saturated	Contains the maximum amount of solute that can dissolve at a given temperature.		No more salt dissolves if added.
Supersaturated	Contains more solute than the maximum amount that can dissolve. It is unstable.		Made by carefully cooling a hot, saturated solution.

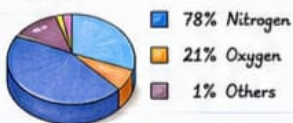
On adding a **seed crystal** or **shaking** → extra solute comes out in form of crystals.

1.15 Concentration Terms

• Dilute Solution	Small amount of solute in large amount of solvent.	
• Concentrated Solution	Large amount of solute in small amount of solvent.	
• Saturated Solution	Contains maximum amount of solute at given temperature.	
• Unsaturated Solution	Contains less than maximum amount of solute.	
• Supersaturated Solution	Contains more than maximum amount of solute (unstable).	

1.16 Some Common Examples

1. Air



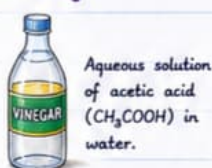
A solution of gases (solid & liquid particles are absent).

2. Brass



A solid solution of zinc in copper.

3. Vinegar



4. Seawater



Contains salts like NaCl , MgCl_2 , KCl dissolved in water.

5. Soft Drinks



CO_2 is dissolved in water along with sugars and flavours.

Quick Comparison

Property	Solution	Suspension	Colloid
Nature	Homogeneous	Heterogeneous	Apparent homogeneous
Particle size	$< 1 \text{ nm}$	$> 1000 \text{ nm}$	$1 \text{ nm} - 1000 \text{ nm}$
Visibility of particles	Not visible	Visible to naked eye	Not visible to naked eye
Stability	Stable	Unstable (settle down)	Relatively stable
Tyndall effect	Absent	May or may not be seen	Present
Filtration	Passes through filter paper	Does not pass through	Passes through filter paper

Memory Trick

To remember concentration terms (from less solute to more):

"**Dil Se Concentrated Tak Saturated Hai, Supersaturated To Extra Badh Gai!**"



Key Takeaways

- ✓ A solution is a homogeneous mixture of two or more substances.
- ✓ Concentration tells how much solute is present.
- ✓ Colligative properties depend only on number of particles.
- ✓ Solubility changes with temperature & pressure.
- ✓ Solutions are everywhere in our daily life!